Package 'fastrerandomize'

October 19, 2026

Title Hardware-Accelerated Rerandomization for Improved Balance

Version 0.4

Description Provides hardware-accelerated tools for performing rerandomization and randomization testing in experimental research. Using a 'JAX' backend, the package enables exact rerandomization inference even for large experiments with hundreds of billions of possible randomizations. Key functionalities include generating pools of acceptable rerandomizations based on covariate balance, conducting exact randomization tests, and performing pre-analysis evaluations to determine optimal rerandomization acceptance thresholds. The package supports various hardware acceleration frameworks including 'CPU', 'CUDA', and 'METAL', making it versatile across accelerated computing environments. This allows researchers to efficiently implement stringent rerandomization designs and conduct valid inference even with large sample sizes. The package is partly based on Jerzak and Goldstein (2023) <doi:10.48550/arXiv.2310.00861>.

URL https://github.com/cjerzak/fastrerandomize-software

BugReports https://github.com/cjerzak/fastrerandomize-software/issues

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`build_backend` *A function to build the environment for fastrerandomize. Builds a conda environment in which 'JAX' and 'np' are installed. Users can also create a conda environment where 'JAX' and 'np' are installed themselves.*

Description

A function to build the environment for fastrerandomize. Builds a conda environment in which 'JAX' and 'np' are installed. Users can also create a conda environment where 'JAX' and 'np' are installed themselves.

Usage

```
build_backend(conda_env = "fastrerandomize_env", conda = "auto")
```

Arguments

`conda_env`	(default = `"fastrerandomize_env"`) Name of the conda environment in which to place the backends.
`conda`	(default = `auto`) The path to a conda executable. Using `"auto"` allows reticulate to attempt to automatically find an appropriate conda binary.

Value

Invisibly returns NULL; this function is used for its side effects of creating and configuring a conda environment for `fastrerandomize`. This function requires an Internet connection. You can find out a list of conda Python paths via: `Sys.which("python")`

Examples

```
## Not run:
# Create a conda environment named "fastrerandomize_env"
# and install the required Python packages (jax, numpy, etc.)
build_backend(conda_env = "fastrerandomize_env", conda = "auto")

# If you want to specify a particular conda path:
# build_backend(conda_env = "fastrerandomize_env", conda = "/usr/local/bin/conda")

## End(Not run)
```

`check_jax_availability`

Check if 'Python' and 'JAX' are available

Description

This function checks if 'Python' and 'JAX' can be accessed via 'reticulate'. If not, it returns 'NULL' and prints a message suggesting to run 'build_backend()'.

Usage

```
check_jax_availability(conda_env = "fastrerandomize_env", conda = "auto")
```

Arguments

`conda_env`	A character string specifying the name of the conda environment. Default is '"fastrerandomize_env"'.
`conda`	The path to a conda executable, or '"auto"'. Default is '"auto"'.

Value

Returns 'TRUE' (invisibly) if both 'Python' and 'JAX' are available; otherwise returns 'NULL'.

Examples

```
## Not run:
  check_jax_availability()

## End(Not run)
```

compute_diff_at_tau_for_oneW_R

Compute potential outcome difference in means for a single assignment under a hypothesized tau in base R

Description

Compute potential outcome difference in means for a single assignment under a hypothesized tau in base R

Usage

```
compute_diff_at_tau_for_oneW_R(Wprime, obsY, obsW, tau)
```

Arguments

Wprime	A 0/1 assignment vector for which to compute the diff in means.
obsY	Observed outcome vector.
obsW	Observed assignment vector.
tau	The hypothesized true effect for the shift in outcomes under treatment.

Value

Scalar difference in means for the assignment Wprime.

diagnose_rerandomization

Diagnostic map from observed (or targeted) balance to precision and stringency

Description

Implements the calculations in Theorem 1 and Appendix D of the paper involving: (1) Realized RMSE from an observed Mahalanobis distance M (or SMDs); (2) Ex-ante RMSE when accepting assignments with M < a (equivalently, with acceptance probability q under complete randomization); (3) largest acceptance probability q that attains a user-specified precision goal, provided via an RMSE target or via a power target (alpha, 1-beta, |tau|).

Usage

```
diagnose_rerandomization(
  smd = NULL,
  M = NULL,
  d = NULL,
  n_T,
  n_C,
  sigma = NULL,
  R2 = NULL,
```

```
  rmse_goal = NULL,
  tau = NULL,
  alpha = 0.05,
  power = 0.8,
  two_sided = TRUE,
  q_min = 1e-09,
  q_tol = 1e-10
)
```

Arguments

smd	Optional numeric vector of standardized mean differences; if supplied, M is computed as sum(smd^2), and d = length(smd).
M	Optional scalar Mahalanobis distance M; if provided without 'smd', you must also supply 'd' (the number of covariates used in M).
d	Optional integer number of covariates (needed if supplying only 'M').
n_T	Integer, number of treated units.
n_C	Integer, number of control units.
sigma	Optional outcome noise SD (sigma). If 'NULL', absolute RMSEs cannot be formed; dimensionless "per-sigma" factors are still returned.
R2	Optional model R^2 for Y ~ X under the linear potential-outcomes model. Must lie in [0,1). If 'NULL', RMSEs that require R^2 are returned as NA, but the "per-sigma" formulas that do not need R^2 are still shown when possible.
rmse_goal	Optional numeric target for RMSE (same units as Y). If supplied (with sigma and R2), the largest q achieving this ex-ante goal is returned.
tau	Optional effect size |tau| (same units as Y) to back out an RMSE goal via a normal approximation to power.
alpha	Size of a two-sided test (default 0.05).
power	Desired power 1 - beta (default 0.80). Used only if 'tau' is given.
two_sided	Logical; if FALSE, uses a one-sided z-threshold for power inversion.
q_min	Lower bound for numerical search over q (default 1e-9).
q_tol	Absolute tolerance for q root-finding (default 1e-10).

Details

Realized (conditional) RMSE: with standardized/whitened X and typical orientation,

$$\mathrm{RMSE}_{\text{realized}} \approx \sqrt{\sigma^2\left(\frac{1}{n_T}+\frac{1}{n_C}\right)+\frac{\sigma^2_{\text{Prog}}}{d}\,M} \;=\; \sigma\sqrt{\left(\frac{1}{n_T}+\frac{1}{n_C}\right)+\frac{R^2}{1-R^2}\,\frac{M}{d}}\,,$$

and the conservative upper bound replaces σ^2_{Prog}/d by σ^2_{Prog}.

Ex-ante (design-stage) RMSE under thresholding: with acceptance rule M <= a (acceptance probability q),

$$\mathbb{E}[\mathrm{MSE} \mid M \le a] = \left(\frac{1}{n_T}+\frac{1}{n_C}\right)\left(\sigma^2 + v_a(d)\,\sigma^2_{\text{Prog}}\right),$$

where $v_a(d) = \Pr(\chi^2_{d+2} \le c)/\Pr(\chi^2_d \le c)$ and $c = a/(\frac{1}{n_T}+\frac{1}{n_C})$. Since $q = \Pr(\chi^2_d \le c)$, we can parameterize by q: $v(q;d) = \Pr(\chi^2_{d+2} \le \chi^2_{d;q})/q$, with $\chi^2_{d;q}$ the q-th quantile.

Power inversion (Appendix D): for two-sided size α and power $1-\beta$, a normal approximation suggests the RMSE goal $|\tau|/(z_{1-\alpha/2}+z_{1-\beta})$.

Value

A list of class '"fastrerandomize_diagnostic"' with elements:

- `inputs`: Echo of parsed inputs and derived quantities (M, d, S=1/n_T+1/n_C).
- `realized`: `rmse_factor` (dimensionless, per sigma), `rmse`, and conservative `rmse_upper_factor`, `rmse_upper`.
- `power_check`: If `tau`, `alpha`, `power`, `sigma`, `R2` are given, includes `z_needed`, `z_realized`, and `already_sufficient`.
- `recommendation`: If a target is supplied (via `rmse_goal` or `tau`), returns `q_star`, `a_star`, `v_star`, `rmse_exante`, `expected_M_accepted`, and `expected_draws_per_accept` = `1/q_star`.

Examples

```
# Example 1: observed SMDs, realized precision only (dimensionless factors)
smd <- c(0.10, -0.05, 0.08, 0.02)  # standardized mean differences
out1 <- diagnose_rerandomization(smd = smd, n_T = 100, n_C = 100)
print(out1)

# Example 2: same, but supply sigma and R^2 for absolute RMSE
out2 <- diagnose_rerandomization(smd = smd, n_T = 100, n_C = 100, sigma = 1.2, R2 = 0.4)

# Example 3: choose q to hit a power target (two-sided alpha=.05, 80% power, |tau|=0.2)
out3 <- diagnose_rerandomization(smd = smd, n_T = 100, n_C = 100, sigma = 1.2, R2 = 0.4,
                tau = 0.2, alpha = 0.05, power = 0.80)

# Analyze rerandomization recommendation given contextual factors
out3$recommendation

# Example 4: choose q to hit an absolute RMSE goal directly
out4 <- diagnose_rerandomization(M = sum(smd^2), d = length(smd), n_T = 100, n_C = 100,
                sigma = 1.2, R2 = 0.4, rmse_goal = 0.25)
```

---

diff_in_means_R *Simple difference in means in base R*

Description

Simple difference in means in base R

Usage

```
diff_in_means_R(Y, W)
```

Arguments

`Y`	Numeric outcome vector.
`W`	0/1 treatment assignment vector.

Value

Scalar difference in means: mean(Y|W=1) - mean(Y|W=0).

fastrerandomize_class *Constructor for fastrerandomize randomizations*

Description

Create an S3 object of class `fastrerandomize_randomizations` that stores the randomizations (and optionally balance statistics) generated by functions such as `generate_randomizations`.

Usage

```
fastrerandomize_class(
  randomizations,
  balance = NULL,
  fastrr_env = NULL,
  call = NULL
)
```

Arguments

`randomizations`	A matrix or array where each row (or slice) represents one randomization.
`balance`	A numeric vector or similar object holding balance statistics for each randomization, or `NULL` if not applicable.
`fastrr_env`	Associated `fastrr_env` environment.
`call`	The function call, if you wish to store it for reference (optional).

Value

An object of class `fastrerandomize_randomizations`.

fastrerandomize_test *Constructor for fastrerandomize randomization test objects*

Description

Constructor for fastrerandomize randomization test objects

Usage

```
fastrerandomize_test(p_value, FI, tau_obs, fastrr_env = NULL, call = NULL, ...)
```

Arguments

`p_value`	A numeric value representing the p-value of the test.
`FI`	A numeric vector (length 2) representing the fiducial interval, or `NULL` if not requested.
`tau_obs`	A numeric value (or vector) representing the estimated treatment effect.
`fastrr_env`	Associated 'fastrr_env' environment.
`call`	An optional function call, stored for reference.
`...`	Other slots you may want to store (e.g. additional diagnostics).

Value

An object of class `fastrerandomize_test`.

`fast_distance` *JAX-accelerated distance calculations*

Description

Compute pairwise distances between the rows of one matrix `A` or two matrices `A` and `B`, using JAX-backed, JIT-compiled kernels. Supports common metrics: Euclidean, squared Euclidean, Manhattan, Chebyshev, Cosine, Minkowski (with optional feature weights), and Mahalanobis (with full or diagonal inverse covariance).

The function automatically batches computations to avoid excessive device memory use.

Usage

```
fast_distance(
  A,
  B = NULL,
 metric = c("euclidean", "sqeuclidean", "manhattan", "chebyshev", "cosine", "minkowski",
    "mahalanobis"),
  p = 2,
  weights = NULL,
  cov_inv = NULL,
  approximate_inv = TRUE,
  squared = FALSE,
  row_batch_size = NULL,
  as_dist = FALSE,
  return_type = "R",
  verbose = FALSE,
  conda_env = "fastrerandomize_env",
  conda_env_required = TRUE
)
```

Arguments

`A`	A numeric matrix with rows as observations and columns as features.
`B`	Optional numeric matrix with the same number of columns as `A`. If `NULL`, distances are computed within `A` (i.e., $n \times n$).
`metric`	Character; one of `"euclidean"`, `"sqeuclidean"`, `"manhattan"`, `"chebyshev"`, `"cosine"`, `"minkowski"`, `"mahalanobis"`. Default is `"euclidean"`.
`p`	Numeric order for Minkowski distance (must be > 0). Default is `2`.
`weights`	Optional numeric vector of length `ncol(A)` with nonnegative feature weights. Used for `"minkowski"` and `"manhattan"` (the latter is equivalent to Minkowski with `p = 1`).
`cov_inv`	Optional inverse covariance matrix (p x p) for Mahalanobis (ignored if `approximate_inv = TRUE`). If not supplied and `approximate_inv = FALSE`, the covariance of `rbind(A, B)` is estimated and (pseudo-)inverted in double precision.

approximate_inv
: Logical; if TRUE and metric = "mahalanobis", uses a diagonal inverse (reciprocal variances) for speed and robustness. Default TRUE.

squared
: Logical; if TRUE, return squared distances when supported ("euclidean" and "mahalanobis"). Ignored for other metrics. Default FALSE.

row_batch_size
: Optional integer; number of rows of A to process per batch. If NULL, a safe size is chosen automatically.

as_dist
: Logical; if TRUE and B is NULL, return a base dist object (for symmetric metrics). Default FALSE.

return_type
: Either "R" (convert to base R matrix/dist) or "jax" (return a JAX array). Default "R".

verbose
: Logical; print batching progress. Default FALSE.

conda_env
: Character; conda environment name used by reticulate. Default "fastrerandomize_env".

conda_env_required
: Logical; whether the specified conda environment must be used. Default TRUE.

Details

- **Mahalanobis**: with approximate_inv = TRUE, the diagonal of the pooled covariance is used (variance stabilizer); otherwise a full inverse covariance is used. - **Weighted distances**: supply weights (length p) for "minkowski" and "manhattan" (the latter uses p = 1). - Euclidean, squared Euclidean and Mahalanobis distances are computed as $\|a\|^2 + \|b\|^2 - 2a^\top b$ (a matrix product) on inputs that are centered, and scaled or whitened for Mahalanobis, in double precision first. - Computations run in float32 and are JIT-compiled with JAX; where applicable, GPU/Metal/CPU device selection follows your existing backend.

Value

An $n \times m$ distance matrix in the format specified by return_type. If as_dist = TRUE and B = NULL (symmetric case), returns a dist object.

See Also

dist

Examples

```
## Not run:
# Simple Euclidean within-matrix distances (returns an n x n matrix)
X <- matrix(rnorm(50 * 8), 50, 8)
D <- fast_distance(X, metric = "euclidean")

# Cosine distance between two sets
A <- matrix(rnorm(100 * 16), 100, 16)
B <- matrix(rnorm(120 * 16), 120, 16)
Dcos <- fast_distance(A, B, metric = "cosine")

# Minkowski with p = 3 and feature weights
w <- runif(ncol(A))
Dm3 <- fast_distance(A, B, metric = "minkowski", p = 3, weights = w)

# Mahalanobis (diagonal approx, fast & robust)
Dmah_diag <- fast_distance(X, metric = "mahalanobis", approximate_inv = TRUE)
```

```
# Mahalanobis with full inverse (computed internally)
Dmah_full <- fast_distance(X, metric = "mahalanobis", approximate_inv = FALSE)

# Return a base R 'dist' object
D_dist <- fast_distance(X, metric = "euclidean", as_dist = TRUE)

## End(Not run)
```

`find_fiducial_interval_R` *Fiducial interval logic in base R, for randomization test*

Description

Fiducial interval logic in base R, for randomization test

Usage

```
find_fiducial_interval_R(
  obsW,
  obsY,
  allW,
  tau_obs,
  alpha = 0.05,
  c_initial = 2,
  n_search_attempts = 500
)
```

Arguments

`obsW`	Observed assignment (0/1).
`obsY`	Observed outcome.
`allW`	Matrix of candidate random assignments (rows = assignments).
`tau_obs`	Observed difference in means with obsW, obsY.
`alpha`	Significance level (default 0.05).
`c_initial`	A finite positive step-size scale (default 2). The default preserves the standard update scale.
`n_search_attempts`	Number of bracket search attempts (default 500).

Value

2-element numeric vector [lower, upper] or [NA, NA] if none accepted.

`generate_randomizations` *Generate randomizations for a rerandomization-based experimental design*

Description

This function generates randomizations for experimental design using either exact enumeration or Monte Carlo sampling methods. It provides a unified interface to both approaches while handling memory and computational constraints appropriately.

Usage

```
generate_randomizations(
  n_units,
  n_treated,
  X = NULL,
  randomization_accept_prob,
  threshold_func = NULL,
  max_draws = 10^6,
  batch_size = NULL,
  randomization_type = "monte_carlo",
  approximate_inv = TRUE,
  file = NULL,
  return_type = "R",
  verbose = TRUE,
  conda_env = "fastrerandomize_env",
  conda_env_required = TRUE,
  seed = NULL
)
```

Arguments

`n_units`	An integer specifying the total number of experimental units.
`n_treated`	An integer specifying the number of units to be assigned to treatment.
`X`	A numeric matrix of covariates used for balance checking. For exact enumeration, `NULL` returns every assignment without balance filtering; Monte Carlo sampling requires a matrix.
`randomization_accept_prob`	A numeric value between 0 and 1 specifying the probability threshold for accepting randomizations based on balance.
`threshold_func`	A 'JAX' function that computes a balance measure for each randomization (see `generate_randomizations_mc` for its signature). Default (`NULL`) uses Hotelling's T-squared statistic.
`max_draws`	An integer specifying the maximum number of randomizations to draw in Monte Carlo sampling.
`batch_size`	An integer specifying batch size for Monte Carlo processing, or `NULL` (default) to size batches automatically. Results do not depend on the batch size.

`randomization_type`
: A string specifying the type of randomization: either `"exact"` or `"monte_carlo"`.

`approximate_inv`
: A logical value indicating whether to use an approximate inverse (diagonal of the covariance matrix) instead of the full matrix inverse when computing balance metrics. This can speed up computations for high-dimensional covariates. Default is `TRUE`.

`file`
: Optional path for a CSV containing assignment columns `W1`, `W2`, ..., and an optional final `balance` column. When supplied, the CSV is written instead of returning the in-memory S3 object.

`return_type`
: A string specifying the format of the returned randomizations and balance measures. Allowed values are `"R"` for base R objects (e.g., `matrix`, `numeric`) or `"jax"` for 'JAX' arrays. Default is `"R"`.

`verbose`
: A logical value indicating whether to print progress information. Default is `TRUE`.

`conda_env`
: A character string specifying the name of the conda environment to use via `reticulate`. Default is `"fastrerandomize_env"`.

`conda_env_required`
: A logical indicating whether the specified conda environment must be strictly used. If `TRUE`, an error is thrown if the environment is not found. Default is `TRUE`.

`seed`
: Optional whole-number seed for Monte Carlo sampling. If `NULL`, a seed is drawn from R's full positive integer range. Ignored for exact enumeration. Default is `NULL`.

Details

The function supports two methods of generating randomizations:

1. Exact enumeration: Generates all possible randomizations (memory intensive but exact).
2. Monte Carlo sampling: Generates randomizations through sampling (more memory efficient).

For large problems (e.g., X with >20 rows), Monte Carlo sampling is recommended.

Value

When `file = NULL`, returns an S3 object with slots:

- `randomizations` An array where each row represents one accepted treatment assignment.
- `balance` A numeric vector containing the corresponding balance measures, or `NULL` when balance filtering is not performed.
- `fastrr_env` The fastrerandomize environment.
- `call` The matched function call.

When `file` is supplied, invisibly returns the normalized CSV path.

See Also

`generate_randomizations_exact` for the exact enumeration method. `generate_randomizations_mc` for the Monte Carlo sampling method.

Examples

```
## Not run:
# Generate synthetic data
X <- matrix(rnorm(20*5), 20, 5)

# Generate randomizations using exact enumeration
RandomizationSet_Exact <- generate_randomizations(
               n_units = nrow(X),
               n_treated = round(nrow(X)/2),
               X = X,
               randomization_accept_prob=0.1,
               randomization_type="exact")

# Generate randomizations using Monte Carlo sampling
RandomizationSet_MC <- generate_randomizations(
               n_units = nrow(X),
               n_treated = round(nrow(X)/2),
               X = X,
               randomization_accept_prob = 0.1,
               randomization_type = "monte_carlo",
               max_draws = 100000,
               batch_size = 1000)

## End(Not run)
```

---

generate_randomizations_exact

Generate Complete Randomizations with Optional Balance Constraints

---

Description

Generates all possible treatment assignments for a completely randomized experiment, optionally filtering them based on covariate balance criteria. The function can generate either all possible randomizations or a subset that meets specified balance thresholds using Hotelling's T-squared statistic.

Usage

```
generate_randomizations_exact(
  n_units,
  n_treated,
  X = NULL,
  randomization_accept_prob = 1,
  approximate_inv = TRUE,
  threshold_func = NULL,
  verbose = TRUE,
  conda_env = "fastrerandomize_env",
  conda_env_required = TRUE
)
```

Arguments

n_units	An integer specifying the total number of experimental units
n_treated	An integer specifying the number of units to be assigned to treatment
X	A numeric matrix of covariates where rows represent units and columns represent different covariates. Default is NULL, in which case all possible randomizations are returned without balance filtering.
randomization_accept_prob	A numeric value between 0 and 1 specifying the quantile threshold for accepting randomizations based on balance statistics. Default is 1 (accept all randomizations).
approximate_inv	A logical value indicating whether to use an approximate inverse (diagonal of the covariance matrix) instead of the full matrix inverse when computing balance metrics. This can speed up computations for high-dimensional covariates. Default is TRUE.
threshold_func	A function that calculates balance statistics for candidate randomizations. Default is VectorizedFastHotel2T2 which computes Hotelling's T-squared statistic.
verbose	A logical value indicating whether to print progress information. Default is TRUE.
conda_env	A character string specifying the name of the conda environment to use via reticulate. Default is "fastrerandomize_env".
conda_env_required	A logical indicating whether the specified conda environment must be strictly used. If TRUE, an error is thrown if the environment is not found. Default is TRUE.

Details

The function works in two main steps: 1. Generates all possible combinations of treatment assignments given n_units and n_treated 2. If covariates (X) are provided, filters these combinations based on balance criteria using the specified threshold function

A warning is emitted before enumeration when the number of combinations is greater than one million. Small exact designs are still enumerated; when balance filtering nominally retains fewer than ten assignments, the function warns rather than failing.

The balance filtering process uses Hotelling's T-squared statistic by default to measure multivariate balance between treatment and control groups. Randomizations are accepted if their balance measure is below the specified quantile threshold.

Value

The function returns a *list* with two elements: candidate_randomizations: an array of randomization vectors M_candidate_randomizations: an array of their balance measures.

Note

This function requires 'JAX' and 'NumPy' to be installed and accessible through the reticulate package.

References

Hotelling, H. (1931). The generalization of Student's ratio. The Annals of Mathematical Statistics, 2(3), 360-378.

See Also

generate_randomizations for full randomization generation function. generate_randomizations_mc for the Monte Carlo version.

Examples

```
## Not run:
# Generate synthetic data
X <- matrix(rnorm(60), nrow = 10)  # 10 units, 6 covariates

# Generate balanced randomizations with covariates
BalancedRandomizations <- generate_randomizations_exact(
  n_units = 10,
  n_treated = 5,
  X = X,
  randomization_accept_prob = 0.25  # Keep top 25% most balanced
)

## End(Not run)
```

---

generate_randomizations_mc

Draws a random sample of acceptable randomizations from all possible complete randomizations using Monte Carlo sampling

---

Description

This function performs sampling with replacement to generate randomizations in a memory-efficient way. It processes randomizations in batches to avoid memory issues and filters them based on covariate balance. The function uses JAX for fast computation and memory management.

Usage

```
generate_randomizations_mc(
  n_units,
  n_treated,
  X,
  randomization_accept_prob = 1,
  threshold_func = NULL,
  max_draws = 1e+05,
  batch_size = NULL,
  approximate_inv = TRUE,
  verbose = TRUE,
  conda_env = "fastrerandomize_env",
  conda_env_required = TRUE,
  seed = NULL
)
```

Arguments

n_units	An integer specifying the total number of experimental units.
n_treated	An integer specifying the number of units to be assigned to treatment.
X	A numeric matrix of covariates used for balance checking. Cannot be NULL.
randomization_accept_prob	A numeric value between 0 and 1 specifying the probability threshold for accepting randomizations based on balance. Default is 1
threshold_func	A traceable JAX function that computes a balance measure for each randomization in a batch. It is called as threshold_func(X, S_inv, S_inv_approx, W, n0, n1, approximate_inv), where X is the covariate matrix, S_inv the inverse sample covariance, S_inv_approx the reciprocal covariate variances, W a batch of 0/1 assignments (one per row), n0/n1 the group sizes, and approximate_inv the logical argument below; it must return one value per row of W. Default (NULL) uses Hotelling's T-squared statistic.
max_draws	An integer specifying the maximum number of randomizations to draw.
batch_size	An integer specifying how many randomizations to process at once, or NULL (default) to size batches automatically from n_units. Lower values use less memory but may be slower. Results do not depend on the batch size.
approximate_inv	A logical value indicating whether to use an approximate inverse (diagonal of the covariance matrix) instead of the full matrix inverse when computing balance metrics. This can speed up computations for high-dimensional covariates. Default is TRUE.
verbose	A logical value indicating whether to print detailed information about batch processing progress, and GPU memory usage. Default is FALSE.
conda_env	A character string specifying the name of the conda environment to use via reticulate. Default is "fastrerandomize_env".
conda_env_required	A logical indicating whether the specified conda environment must be strictly used. If TRUE, an error is thrown if the environment is not found. Default is TRUE.
seed	Optional whole-number seed for the JAX random stream. If NULL, a seed is drawn from R's full positive integer range, so set.seed() can still be used to reproduce the draw. Default is NULL.

Details

The function works by:

1. Deriving the key of draw i as jax.random.split(key, max_draws)[i] inside each batch, without materializing all keys.
2. Permuting the base assignment for each key and computing its balance measure.
3. Keeping the ceiling(max_draws * randomization_accept_prob) most balanced draws and regenerating their assignments from their keys.

All batches run inside one compiled JAX program (on Metal, or when options(fastrerandomize.mc_fused = FALSE) is set, batches are dispatched from R). Covariates are centered and scaled or whitened in double precision before they are sent to the device; constant covariates are dropped with a warning. Every generated assignment is checked to contain exactly n_treated treated units.

Value

The function returns a *list* with three elements: `candidate_randomizations`: an int8 array of randomization vectors; `keys_candidate_randomizations`: their JAX random keys; `M_candidate_randomizations`: an array of their balance measures.

See Also

`generate_randomizations` for full randomization generation function. `generate_randomizations_exact` for the exact version.

Examples

```
## Not run:
# Generate synthetic data
X <- matrix(rnorm(100*5), 100, 5) # 5 covariates

# Generate 1000 randomizations for 100 units with 50 treated
rand_less_strict <- generate_randomizations_mc(
               n_units = 100,
               n_treated = 50,
               X = X,
               randomization_accept_prob=0.01,
               max_draws = 100000,
               batch_size = 1000)

# Use a stricter balance criterion
rand_more_strict <- generate_randomizations_mc(
               n_units = 100,
               n_treated = 50,
               X = X,
               randomization_accept_prob=0.001,
               max_draws = 1000000,
               batch_size = 1000)

## End(Not run)
```

---

generate_randomizations_R

Generate randomizations in base R, filtering by Hotelling's T^2 acceptance

---

Description

Base R function to either do exact enumeration or Monte Carlo random permutations, then keep the fraction whose T^2 is below the acceptance cutoff.

Usage

```
generate_randomizations_R(
  n_units,
  n_treated,
  X,
```

```
  accept_prob,
  random_type,
  max_draws,
  batch_size
)
```

Arguments

n_units	Integer, total number of units.
n_treated	Integer, number of units to be assigned to treatment.
X	Covariate matrix (n_units x p).
accept_prob	Numeric in [0, 1]: keep the fraction of randomizations that have the lowest T^2 up to this quantile.
random_type	Either "exact" or "monte_carlo".
max_draws	If ‘random_type="monte_carlo"‘, how many permutations to sample.
batch_size	If ‘random_type="monte_carlo"‘, how many permutations to handle per chunk.

Value

A list with:

- randomizations: a matrix (rows = accepted assignments).
- balance: numeric vector of T^2 values for each accepted assignment.

---

hotellingT2_R *Compute Hotelling's T-squared statistic in base R*

Description

This function provides a base R implementation of Hotelling's T-squared balance measure, renamed with ‘_R‘ for clarity that it is the R-based analog to the JAX version in fastrerandomize.

Usage

```
hotellingT2_R(X, W)
```

Arguments

X	A numeric matrix of covariates (observations in rows).
W	A 0/1 treatment assignment vector of the same length as nrow(X).

Value

A numeric scalar: Hotelling's T-squared statistic for that assignment.

plot.fastrerandomize_randomizations

Plot method for fastrerandomize_randomizations objects

Description

Generates a histogram of the balance measures for accepted randomizations.

Usage

```
## S3 method for class 'fastrerandomize_randomizations'
plot(x, ...)
```

Arguments

x	An object of class fastrerandomize_randomizations.
...	Further graphical parameters passed to plot.

Value

No return value. This function is called for the side effect of generating a histogram of the accepted balance measures of object with class fastrerandomize_randomizations.

plot.fastrerandomize_test

Plot method for fastrerandomize_test objects

Description

Plots a simple visualization of the observed effect and the fiducial interval (if present) on a horizontal axis.

Usage

```
## S3 method for class 'fastrerandomize_test'
plot(x, ...)
```

Arguments

x	An object of class fastrerandomize_test.
...	Further graphical parameters passed to plot.

Value

No output returned. Performs side effect of plotting fastrerandomize_test class objects.

print.fastrerandomize_randomizations *Print method for fastrerandomize_randomizations objects*

Description

Print method for fastrerandomize_randomizations objects

Usage

```
## S3 method for class 'fastrerandomize_randomizations'
print(x, ...)
```

Arguments

x	An object of class fastrerandomize_randomizations.
...	Further arguments passed to or from other methods.

Value

Prints an object of class fastrerandomize_randomizations.

print.fastrerandomize_test *Print method for fastrerandomize_test objects*

Description

Print method for fastrerandomize_test objects

Usage

```
## S3 method for class 'fastrerandomize_test'
print(x, ...)
```

Arguments

x	An object of class fastrerandomize_test.
...	Further arguments passed to or from other methods.

Value

No return value, prints object of class fastrerandomize_test.

print2 *Print timestamped messages with optional quieting*

Description

This function prints messages prefixed with the current timestamp in a standardized format. Messages can be suppressed using the quiet parameter.

Usage

```
print2(text, quiet = FALSE)
```

Arguments

text	A character string containing the message to be printed.
quiet	A logical value indicating whether to suppress output. Default is FALSE.

Details

The function prepends the current timestamp in "YYYY-MM-DD HH:MM:SS" format to the provided message.

Value

No return value, called for side effect of printing with timestamp.

See Also

Sys.time for the underlying timestamp functionality.

Examples

```
# Print a basic message with timestamp
print2("Processing started")

# Suppress output
print2("This won't show", quiet = TRUE)

# Use in a loop
for(i in 1:3) {
  print2(sprintf("Processing item %d", i))
}
```

QJEData *QJEData: Agricultural Treatment Experiment Data*

Description

Data from a field experiment studying moral hazard in tenancy contracts in agriculture.

After subsetting, this dataset includes observations on 968 experimental units with the following variables of interest: household composition, treatment assignment, and agricultural outcomes.

Usage

```
data(QJEData)
```

Format

A data frame with 968 rows and 7 columns:

children Numeric (integer). Number of children in the household. Larger numbers may reflect increased household labor needs and different investment or effort incentives.

married Numeric/binary. Whether the household head is currently married (1) or not (0). Marital status may influence decision-making and risk preferences in farming.

hh_size Numeric (integer). Household size. Differences in family labor availability or consumption needs can influence effort levels and thus relate to moral hazard in production decisions.

hh_sexrat Numeric. The ratio of adult men to adult women in the household. Imbalances in the male–female ratio can affect labor division and investment decisions.

treat1 Numeric/binary. Primary treatment indicator (e.g., whether a farmer is offered a specific tenancy contract or cost-sharing arrangement).

R_yield_ELA_sqm Numeric. Crop yield per square meter (e.g., kilograms of output per square meter). This is a principal outcome measure for evaluating productivity and treatment impact on farm performance.

ELA_Fertil_D Numeric/binary. Indicator for whether fertilizer was used (1) or not (0). This measures input investment—a key mechanism in moral hazard models (farmers may alter input use under different contracts).

Source

Burchardi, K.B., et al. (2019). Moral hazard: Experimental evidence from tenancy contracts. *The Quarterly Journal of Economics*, 134(1), 281-347.

`randomization_test` *Fast randomization test*

Description

Fast randomization test

Usage

```
randomization_test(
  obsW = NULL,
  obsY = NULL,
  alpha = 0.05,
  candidate_randomizations = NULL,
  candidate_randomizations_array = NULL,
  n0_array = NULL,
  n1_array = NULL,
  findFI = FALSE,
  c_initial = 2,
  conda_env = "fastrerandomize_env",
  conda_env_required = TRUE
)
```

Arguments

`obsW`	A numeric vector where `0`'s correspond to control units and `1`'s to treated units.
`obsY`	A required numeric vector of observed outcomes with the same length as `obsW`.
`alpha`	The significance level for the test. Default is `0.05`.
`candidate_randomizations`	A numeric matrix of candidate randomizations.
`candidate_randomizations_array`	An optional 'JAX' array of candidate randomizations. If not provided, the function coerces `candidate_randomizations` into a 'JAX' array.
`n0_array`	An optional array specifying the number of control units.
`n1_array`	An optional array specifying the number of treated units.
`findFI`	A logical value indicating whether to find the fiducial interval. Default is FALSE.
`c_initial`	A finite positive step-size scale for the fiducial interval search. The default of `2` preserves the standard update scale; smaller or larger values decrease or increase the stochastic update size.
`conda_env`	A character string specifying the name of the conda environment to use via `reticulate`. Default is `"fastrerandomize_env"`.
`conda_env_required`	A logical indicating whether the specified conda environment must be strictly used. If `TRUE`, an error is thrown if the environment is not found. Default is `TRUE`.

Details

All statistics are computed from one pass over the candidates: for a binary candidate `w`, the difference in means under the sharp null of effect `tau` equals `a_w + tau * (c_w - b_w)`, where `a_w` is the difference in means of `obsY` under `w`, `b_w` that of `obsW`, and `c_w = sum(w) / n1`. The fiducial search then runs on these per-candidate values. `tau_obs` is computed in double precision, and candidates whose statistic ties `tau_obs` up to float32 error are re-evaluated in double precision, so exact ties are counted consistently.

Value

Returns an S3 object with slots:

- `p_value` A numeric value or vector representing the p-value of the test (or the expected p-value under the prior structure specified in the function inputs).
- `FI` A numeric vector representing the fiducial interval if `findFI=TRUE`.
- `tau_obs` A numeric value or vector representing the estimated treatment effect(s).
- `fastrr_env` The fastrerandomize environment.
- `candidate_randomizations` The candidate matrix, when supplied as an R matrix; otherwise `NULL`, and the 'JAX' array is kept in `candidate_randomizations_array` (it is not copied back to R).

References

- Zhang, Y. and Zhao, Q., 2023. What is a randomization test?. Journal of the American Statistical Association, 118(544), pp.2928-2942.

See Also

`generate_randomizations` for randomization generation function.

Examples

```
## Not run:
# A small synthetic demonstration with 6 units, 3 treated and 3 controls:

# Generate pre-treatment covariates
X <- matrix(rnorm(24*2), ncol = 2)

# Generate candidate randomizations
RandomizationSet_MC <- generate_randomizations(
  n_units = nrow(X),
  n_treated = round(nrow(X)/2),
  X = X,
  randomization_accept_prob = 0.1,
  randomization_type = "monte_carlo",
  max_draws = 100000,
  batch_size = 1000
)

# Generate outcome
W <- RandomizationSet_MC$randomizations[1,]
obsY <- rnorm(nrow(X), mean = 2 * W)

# Perform randomization test
```

```
results_base <- randomization_test(
  obsW = W,
  obsY = obsY,
  candidate_randomizations = RandomizationSet_MC$randomizations
)
print(results_base)

# Perform randomization test with fiducial interval
result_fi <- randomization_test(
  obsW = W,
  obsY = obsY,
  candidate_randomizations = RandomizationSet_MC$randomizations,
  findFI = TRUE
)
print(result_fi)

## End(Not run)
```

`randomization_test_R` *Base R randomization test: difference in means + optional fiducial interval*

Description

Base R randomization test: difference in means + optional fiducial interval

Usage

```
randomization_test_R(obsW, obsY, allW, findFI = FALSE, alpha = 0.05)
```

Arguments

`obsW`	Observed assignment (0/1).
`obsY`	Observed outcome vector.
`allW`	Matrix of candidate random assignments (rows = assignments).
`findFI`	Logical, whether to compute fiducial interval as well.
`alpha`	Significance level (default 0.05).

Value

A list with p_value, tau_obs, and (optionally) FI if ‘findFI=TRUE‘.

summary.fastrerandomize_randomizations *Summary method for fastrerandomize_randomizations objects*

Description

Summary method for fastrerandomize_randomizations objects

Usage

```
## S3 method for class 'fastrerandomize_randomizations'
summary(object, ...)
```

Arguments

object	An object of class fastrerandomize_randomizations.
...	Further arguments passed to or from other methods.

Value

A list with summary statistics, printed by default.

summary.fastrerandomize_test *Summary method for fastrerandomize_test objects*

Description

Summary method for fastrerandomize_test objects

Usage

```
## S3 method for class 'fastrerandomize_test'
summary(object, ...)
```

Arguments

object	An object of class fastrerandomize_test.
...	Further arguments passed to or from other methods.

Value

Returns an (invisible) list with a summary of fastrerandomize_test class objects.

YOPData *YOPData*

Description

Data from a re-analysis of the Youth Opportunities Program anti-poverty RCT in Uganda, with satellite imagery neural representations linked to RCT units.

Usage

```
data(YOPData)
```

Format

A list containing two data frames:

RCTData Treatment, outcome, and geolocation information

ImageEmbeddings CLIP-RSICD neural embeddings of satellite imagery

Source

- Blattman, C., Fiala, N. and Martinez, S. (2020). The Long-term Impacts of Grants on Poverty: Nine-year Evidence from Uganda's Youth Opportunities Program. American Economic Review: Insights, 2(3), 287-304.
- Jerzak, C.T., Johansson, F.D. and Daoud, A. (2023). Image-based Treatment Effect Heterogeneity. Conference on Causal Learning and Reasoning, 531-552. PMLR.

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